

Half Adder

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

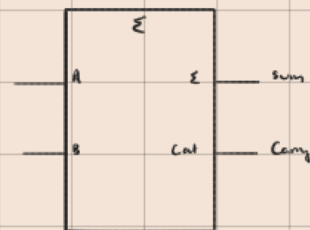


- 2 inputs , 2 outputs

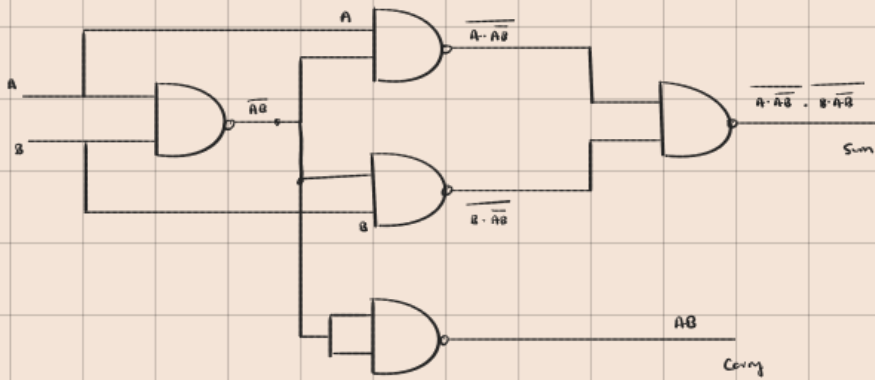
$$S = \bar{A}B + \bar{B}A$$

$$C = AB$$

$$S = A \oplus B$$

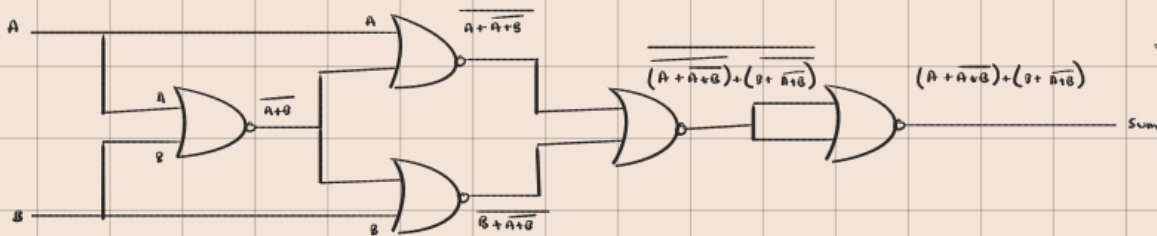


Half Adder Using NAND Gates



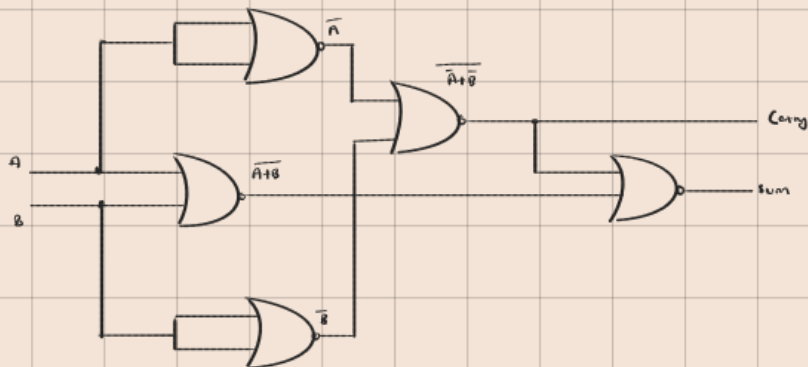
$$\begin{aligned}
 \overline{A \cdot \overline{A} B} &= \overline{B \cdot \overline{A} \overline{B}} \\
 &= (\overline{A \cdot \overline{A} B}) + (\overline{B \cdot \overline{A} \overline{B}}) \\
 &= A(\overline{A} + \overline{B}) + B(\overline{A} + \overline{B}) \\
 &= (\overline{A} \cdot \overline{A}) + (\overline{A} \cdot \overline{B}) + (\overline{B} \cdot \overline{A}) + (\overline{B} \cdot \overline{B}) \\
 &= A\overline{B} + B\overline{A} \\
 &= A \oplus B
 \end{aligned}$$

Half Adder Using NOR Gate



XOR Gate

OR



$$\begin{aligned}
 C &= \overline{\overline{A} + \overline{B}} \\
 &= A \cdot B
 \end{aligned}$$

$$S = \overline{\overline{A} + \overline{B}} + \overline{A + B}$$

$$S = \overline{\overline{A} + \overline{B}} + \overline{A + B}$$

$$S = (\overline{A} + \overline{B}) \cdot (A + B)$$

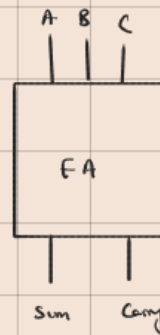
$$= \overline{A}A + \overline{A}B + \overline{B}A + \overline{B}\overline{B}$$

$$= A\overline{B} + \overline{B}A$$

$$= A \oplus B$$

Full Adder

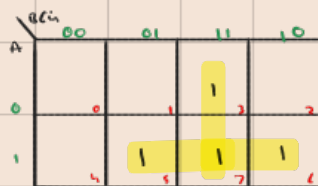
A	B	C _{in}	Sum	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$\begin{aligned}
 S &= \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} \\
 &= \bar{A}(\bar{B}C_{in} + B\bar{C}_{in}) + A(\bar{B}\bar{C}_{in} + BC_{in}) \\
 &= \bar{A}(B \oplus C_{in}) + A(\overline{B \oplus C_{in}})
 \end{aligned}$$

$$S = A \oplus B \oplus C_{in}$$

$$C = \bar{A}B C_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$$

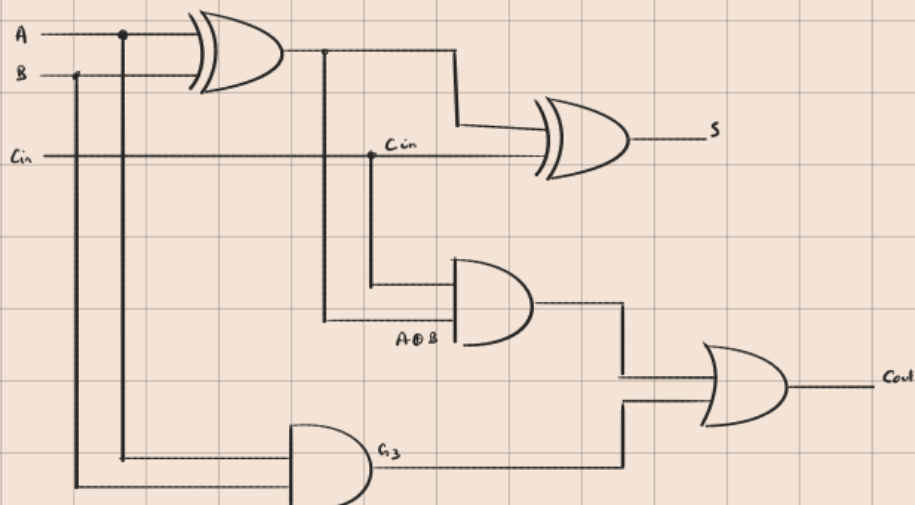


$$C = \bar{A}B C_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$$

$$C = AB(\bar{C}_{in} + C_{in}) + C_{in}(\bar{A}B + A\bar{B})$$

$$C = AB + C_{in}(A \oplus B)$$

$$C = AB + BC_{in} + AC_{in}$$



Full Adder Using Nand

